

```
!pip install statsmodels -q

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import statsmodels.formula.api as smf

print("All libraries loaded!")
```

All libraries loaded!

```
from google.colab import files
uploaded = files.upload()
```

[Choose Files](#) 85334965-...72f84 (1).csv
85334965-5736-457a-b8d4-a077e6872f84 (1).csv(text/csv) - 181277 bytes, last modified: 6/2/2026 - 100% done
Saving 85334965-5736-457a-b8d4-a077e6872f84 (1).csv to 85334965-5736-457a-b8d4-a077e6872f84 (1).csv

```
df = pd.read_csv('85334965-5736-457a-b8d4-a077e6872f84.csv')
print(df.shape)
print(df.head())
print(df.describe())
print(df.isnull().sum())
```

FileNotFoundError Traceback (most recent call last)
[/tmp/ipykernel_5297/119332318.py](#) in <cell line: 0>()
----> 1 df = pd.read_csv('85334965-5736-457a-b8d4-a077e6872f84.csv')
 2 print(df.shape)
 3 print(df.head())
 4 print(df.describe())
 5 print(df.isnull().sum())

----- 4 frames -----
[/usr/local/lib/python3.12/dist-packages/pandas/io/common.py](#) in get_handle(path_or_buf, mode, encoding, compression, memory_map, is_text, errors, storage_options)
 871 if ioargs.encoding and "b" not in ioargs.mode:
 872 # Encoding
--> 873 handle = open(
 874 handle,
 875 ioargs.mode,

FileNotFoundError: [Errno 2] No such file or directory: '85334965-5736-457a-b8d4-a077e6872f84.csv'

Next steps: [Explain error](#)

```
import os
print(os.listdir('/content'))

['.config', '85334965-5736-457a-b8d4-a077e6872f84 (1).csv', 'sample_data']
```

```
df = pd.read_csv('/content/85334965-5736-457a-b8d4-a077e6872f84 (1).csv')
print(df.shape)
print(df.head())
print(df.describe())
print(df.isnull().sum())
```

(4572, 4)				
	TV	Radio	Social_Media	Sales
0	16.0	6.566231	2.907983	54.732757
1	13.0	9.237765	2.409567	46.677897
2	41.0	15.886446	2.913410	150.177829
3	83.0	30.020028	6.922304	298.246340
4	15.0	8.437408	1.405998	56.594181
	TV	Radio	Social_Media	Sales
count	4562.000000	4568.000000	4566.000000	4566.000000
mean	54.066857	18.160356	3.323956	192.466602
std	26.125054	9.676958	2.212670	93.133092
min	10.000000	0.000684	0.000031	31.199409
25%	32.000000	10.525957	1.527849	112.322882
50%	53.000000	17.859513	3.055565	189.231172
75%	77.000000	25.649730	4.807558	272.507922
max	100.000000	48.871161	13.981662	364.079751

```
TV          10
Radio       4
Social_Media 6
Sales       6
dtype: int64
```

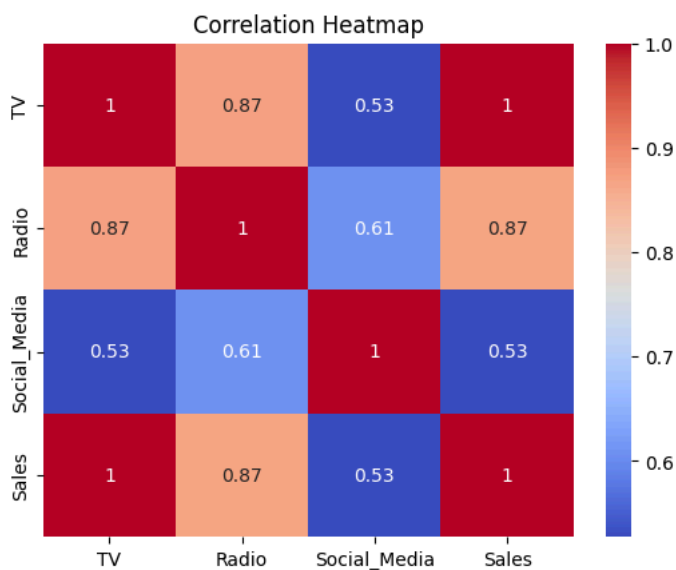
```
# Drop missing values
df = df.dropna()
print("Shape after cleaning:", df.shape)
print(df.isnull().sum())
```

```
Shape after cleaning: (4546, 4)
TV          0
Radio       0
Social_Media 0
Sales       0
dtype: int64
```

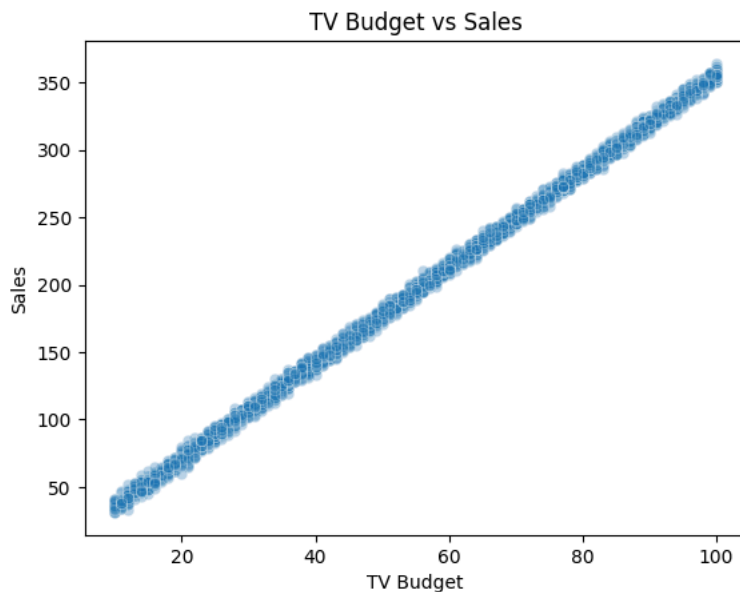
```
# Check correlation with Sales
correlation = df.corr()
print(correlation['Sales'])

# Visualize correlation heatmap
sns.heatmap(correlation, annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

```
TV          0.999497
Radio       0.868638
Social_Media 0.527446
Sales       1.000000
Name: Sales, dtype: float64
```



```
# Scatterplot of TV vs Sales
sns.scatterplot(x='TV', y='Sales', data=df, alpha=0.3)
plt.title('TV Budget vs Sales')
plt.xlabel('TV Budget')
plt.ylabel('Sales')
plt.show()
```



```
# Build OLS regression model
model = smf.ols(formula='Sales ~ TV', data=df).fit()
print(model.summary())
```

```
=====
                        OLS Regression Results
=====
```

Dep. Variable:	Sales	R-squared:	0.999
Model:	OLS	Adj. R-squared:	0.999
Method:	Least Squares	F-statistic:	4.517e+06
Date:	Sun, 14 Jun 2026	Prob (F-statistic):	0.00
Time:	06:25:55	Log-Likelihood:	-11366.
No. Observations:	4546	AIC:	2.274e+04
Df Residuals:	4544	BIC:	2.275e+04
Df Model:	1		
Covariance Type:	nonrobust		

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	coef	std err	t	P> t	[0.025	0.975]
Intercept	-0.1325	0.101	-1.317	0.188	-0.330	0.065
TV	3.5615	0.002	2125.272	0.000	3.558	3.565

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Omnibus:	0.052	Durbin-Watson:	1.998
Prob(Omnibus):	0.974	Jarque-Bera (JB):	0.031
Skew:	-0.001	Prob(JB):	0.985
Kurtosis:	3.012	Cond. No.	138.

```
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```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
# Diagnostic plots
fitted = model.fittedvalues
residuals = model.resid

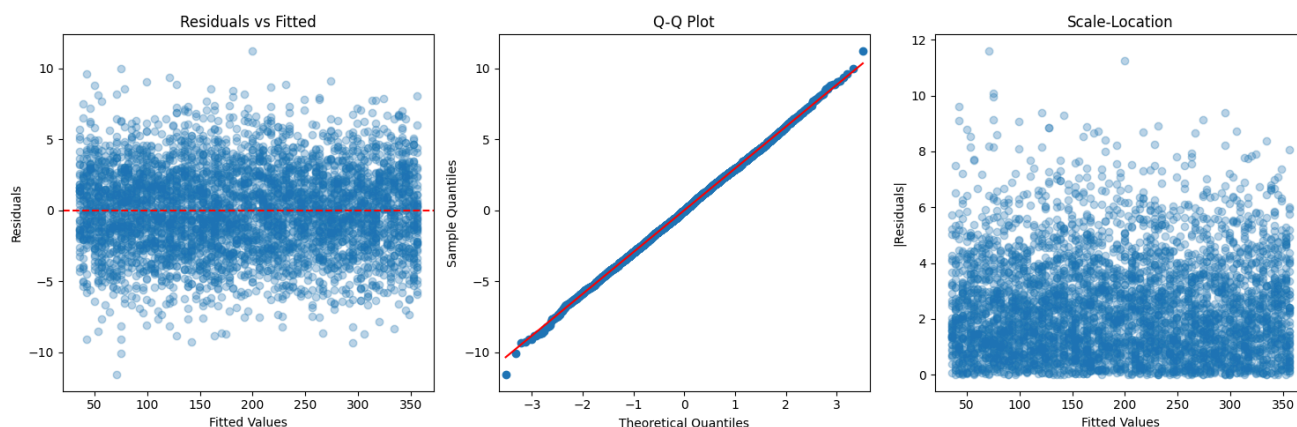
fig, axes = plt.subplots(1, 3, figsize=(15, 5))

# Plot 1: Residuals vs Fitted (Linearity)
axes[0].scatter(fitted, residuals, alpha=0.3)
axes[0].axhline(0, color='red', linestyle='--')
axes[0].set_title('Residuals vs Fitted')
axes[0].set_xlabel('Fitted Values')
axes[0].set_ylabel('Residuals')

# Plot 2: Q-Q Plot (Normality)
sm.qqplot(residuals, line='s', ax=axes[1])
axes[1].set_title('Q-Q Plot')

# Plot 3: Scale-Location (Homoscedasticity)
axes[2].scatter(fitted, abs(residuals), alpha=0.3)
axes[2].set_title('Scale-Location')
axes[2].set_xlabel('Fitted Values')
axes[2].set_ylabel('|Residuals|')
```

```
plt.tight_layout()
plt.show()
```



```
# Business Interpretation
print("=" * 55)
print("MARKETING ROI ANALYSIS - KEY FINDINGS")
print("=" * 55)
print()
print("REGRESSION EQUATION:")
print("Sales = -0.13 + 3.56 * TV")
print()
print("R-SQUARED: 0.999")
print("TV explains 99.9% of the variation in Sales")
print()
print("TV COEFFICIENT: 3.56")
print("Every $1 increase in TV budget generates $3.56 in Sales")
print()
print("P-VALUE: 0.000")
print("TV is statistically significant at 99% confidence level")
print()
print("ASSUMPTION CHECKS:")
print("Linearity      - Residuals randomly scattered around 0")
print("Normality       - Q-Q plot points follow the line closely")
print("Homoscedasticity - Residuals evenly spread across fitted values")
print()
print("RECOMMENDATION:")
print("Allocate majority of marketing budget to TV advertising")
print("TV delivers the strongest ROI at $3.56 per $1 spent")
print("=" * 55)
```

```
=====
MARKETING ROI ANALYSIS - KEY FINDINGS
=====
```

```
REGRESSION EQUATION:
Sales = -0.13 + 3.56 * TV
```

```
R-SQUARED: 0.999
TV explains 99.9% of the variation in Sales
```

```
TV COEFFICIENT: 3.56
Every $1 increase in TV budget generates $3.56 in Sales
```

```
P-VALUE: 0.000
TV is statistically significant at 99% confidence level
```

```
ASSUMPTION CHECKS:
Linearity      - Residuals randomly scattered around 0
Normality       - Q-Q plot points follow the line closely
Homoscedasticity - Residuals evenly spread across fitted values
```

```
RECOMMENDATION:
Allocate majority of marketing budget to TV advertising
```

```
TV delivers the strongest ROI at $3.56 per $1 spent
=====
```

```
Start coding or generate with AI.
```